

# Lesson 51 Practice: Programming Foundations, Compiler and Data Structures

JKSSB Computer Awareness

October 10, 2026

**Provenance:** All 25 questions are original exam-pattern questions (*[Original]*) plus the *[JKSSB FAA-pattern]*, *[JKPSI-pattern]*, *[SSC CGL-pattern]*, and *[JKSSB-pattern]* extension items, including MSQs); none reproduces an official paper item.

**Q1.** *[Original]* Expand OOP.

- A. Object-Oriented Programming
- B. Object-Oriented Processing
- C. Open-Output Programming
- D. Operational Object Programming

**Q2.** *[Original]* Expand LIFO.

- A. Last In First Out
- B. Last In First Open
- C. Large Input Fast Output
- D. Last Input First Order

**Q3.** *[Original]* Which language is the only one a computer executes directly?

- A. Assembly language
- B. Machine language
- C. High-level language
- D. Pseudocode

**Q4.** *[Original]* Assembly language uses short codes such as ADD and MOV, which are called:

- A. mnemonics
- B. keywords
- C. variables
- D. functions

**Q5.** *[Original]* Which translator converts an assembly-language program into machine code?

- A. Compiler
- B. Interpreter

- C. Assembler
- D. Linker

**Q6.** *[Original]* Which statement correctly distinguishes a compiler from an interpreter?

- A. A compiler translates line by line, while an interpreter translates the whole program at once.
- B. A compiler translates the whole program at once, while an interpreter translates and executes it line by line.
- C. Both translate the whole program at once.
- D. A compiler executes source code directly without any translation.

**Q7.** *[Original]* Source code is best described as:

- A. the binary instructions executed directly by the CPU.
- B. the program written by the programmer in a high-level or assembly language.
- C. the machine-readable output produced by a compiler.
- D. firmware stored in read-only memory.

**Q8.** *[Original]* The machine-readable code produced by a compiler from source code is called:

- A. source code
- B. object code
- C. pseudocode
- D. natural-language code

**Q9.** *[Original]* An algorithm is best defined as:

- A. a pictorial representation using standard symbols.
- B. a finite, step-by-step procedure to solve a problem.
- C. the binary code executed by the CPU.
- D. a software tool for finding errors.

**Q10.** *[Original]* In a flowchart, which symbol is used for a decision?

- A. Rectangle
- B. Parallelogram
- C. Diamond
- D. Oval

**Q11.** *[Original]* The process of finding and correcting errors in a program is called:

- A. compiling
- B. debugging
- C. linking
- D. encoding

**Q12.** *[Original]* A program compiles and runs but produces a wrong result. This is an example of a:

- A. syntax error
- B. logical error
- C. runtime error
- D. translation error

**Q13.** *[Original]* In object-oriented programming, a class is:

- A. an instance created from an object.
- B. a blueprint or template from which objects are created.
- C. a permanent secondary-storage location.
- D. a machine-language instruction.

**Q14.** *[Original]* Which OOP feature means “one name, many forms”?

- A. Encapsulation
- B. Inheritance
- C. Polymorphism
- D. Abstraction

**Q15.** *[Original]* A stack follows which order?

- A. FIFO
- B. LIFO
- C. Random
- D. Sorted

**Q16.** *[Original]* In a stack, the insertion and deletion operations are called:

- A. enqueue and dequeue
- B. push and pop
- C. insert and remove
- D. add and delete

**Q17.** *[Original]* A queue follows which order?

- A. LIFO
- B. FIFO
- C. Random
- D. Reverse

**Q18.** *[Original]* Which data structure is hierarchical, consisting of a root node with child nodes?

- A. Array
- B. Stack

- C. Tree
- D. Queue

**Q19.** *[Original]* Which of the following is **NOT** true of a compiler?

- A. It translates the whole program at once.
- B. It produces object code.
- C. It translates and executes the program line by line.
- D. It reports errors after translating the whole program.

**Q20.** *[Original]* Evaluate the statements:

- I. An array stores elements of the same type in contiguous memory locations.
- II. A linked list stores its elements in contiguous memory locations.
- III. A stack follows the Last In, First Out order.

- A. I and II only
- B. II and III only
- C. I and III only
- D. I, II, and III

**Q21.** *[JKSSB FAA-pattern]* A compiler translates a program:

- A. as a whole before execution.
- B. one printed page at a time.
- C. only during paper printing.
- D. without reading the code.

**Q22.** *[JKPSI-pattern]* Which data structure works on a last-in-first-out basis?

- A. Stack
- B. Queue
- C. Printer tray
- D. Hard disk

**Q23.** *[SSC CGL-pattern]* Which of the following is a high-level programming language?

- A. Python
- B. Machine code
- C. Printer ink
- D. Speaker wire

**Q24.** *[JKSSB-pattern]* (**MSQ: select ALL correct options.**) Which are programming languages?

- A. C
- B. Java

- C. Python
  - D. Hard disk
- Q25.** *[SSC CGL-pattern]* (MSQ: select ALL correct options.) Which statements about translators and structures are correct?
- A. Compilers translate whole programs at once.
  - B. Interpreters translate line by line.
  - C. A queue follows first-in-first-out order.
  - D. A stack follows first-in-first-out order.
- Q26.** *[Original]* Which translator converts an entire high-level program before execution?
- A. Compiler
  - B. Interpreter
  - C. Text editor
  - D. Linker only
- Q27.** *[Original]* Which data structure follows the LIFO principle?
- A. Stack
  - B. Queue
  - C. Array only
  - D. Graph
- Q28.** *[Original]* A written step-by-step procedure for solving a problem is an:
- A. Algorithm
  - B. Operating system
  - C. Database record
  - D. Device driver
- Q29.** *[Original]* Which language is closest to the processor's native instruction representation?
- A. Machine language
  - B. High-level language
  - C. Natural language
  - D. Spreadsheet formula language only
- Q30.** *[Original, MSQ drill]* (MSQ: select ALL correct options.) Which pairings are correct?
- A. Stack — LIFO
  - B. Queue — FIFO
  - C. Compiler — translates a program before execution
  - D. Debugging — intentionally adds errors